CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
FACULTY OF SOFTWARE ENGINEERING

SOFTWARE DEVELOPMENT THESIS PROJECT

COURSE: CT250H

DESIGN AND IMPLEMENTATION OF A HUMAN-IN-THE-LOOP AI-ASSISTED RECRUITMENT AUTOMATION PLATFORM FOR CV-JD EVALUATION AND RECOMMENDATION IN IT

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ABSTRACT

Recruitment in the information technology sector requires candidates and recruiters to compare many types of information from curricula vitae and job descriptions. Most job portals support posting, searching, and manual applications, while automated screening tools may not clearly explain how a score is calculated or how an automated action is controlled. This thesis presents CareerFit IT AutoPilot, a web-based recruitment platform that combines a job portal, CV-job description matching, personalized job recommendations, policy-based automation, and Human-in-the-Loop interaction in one workflow.

The system processes candidate profiles, curricula vitae, job descriptions, preferences, and user feedback. Text is normalized and represented with Term Frequency-Inverse Document Frequency vectors. Cosine similarity ranks CV-job pairs and candidate-job recommendations, while the Rocchio relevance-feedback method updates learned representations from explicit feedback. An AutoFit policy checks scores, user consent, thresholds, interaction history, quotas, cooldown rules, time zones, and quiet hours before choosing an action. Email-action links expire, store only hashed tokens, show a confirmation page without changing data, and require a POST request to complete the action. Audit records make actions easier to trace and help prevent a link from being used more than once.

CareerFit is implemented as a role-based web platform for guests, candidates, recruiters, and administrators using a Spring Boot backend, a React and TypeScript frontend, and PostgreSQL persistence managed through Flyway migrations. The refreshed backend suite passed 72 of 72 tests, the production frontend build completed with a largest JavaScript chunk of 378.44 kB, and all 20 Chromium workflow, contract, and resilience tests passed. On a synthetic causal benchmark containing 50 Jobs and 100 CVs, Rocchio increased nDCG@5 from 0.037737 to 0.837737. The final benchmark log contained no optimistic-lock exception, and aggregate runtime health returned HTTP 200 with status UP. These results demonstrate controlled feedback behavior and integrated prototype workflows, not production hiring effectiveness.

The project shows a practical approach to recruitment automation. Explainable scores and configurable policies support users while people remain responsible for

important decisions. Current limitations include lexical vector representations and controlled evaluation data. Semantic representations, larger real-world studies, and stronger management controls are left for future work.
Keywords: recruitment automation; CV–job matching; recommendation system; Human-in-the-Loop; Rocchio feedback; explainable automation.

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